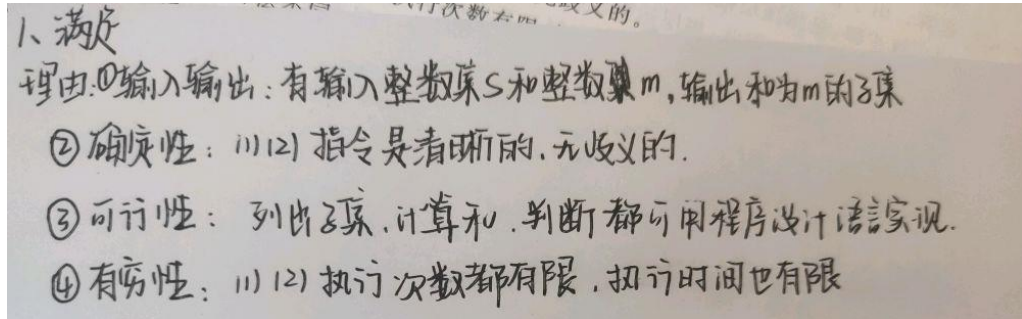
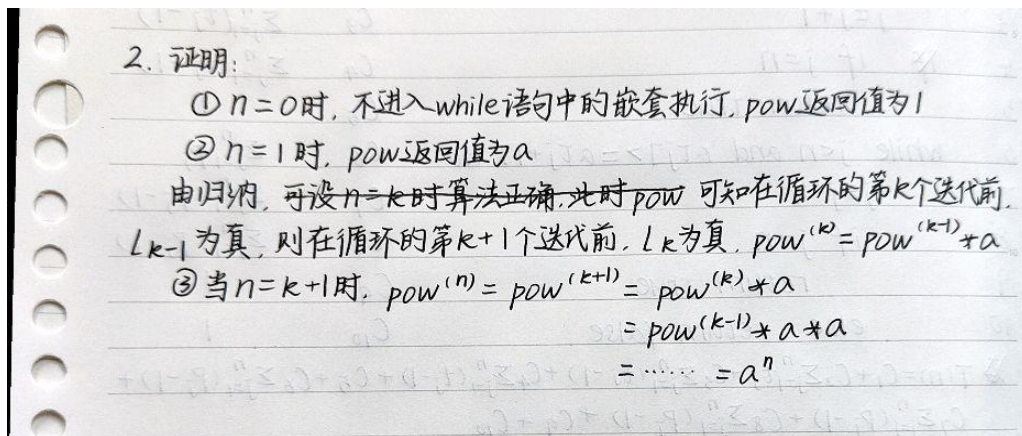


算法 (AI) 第一周优秀作业

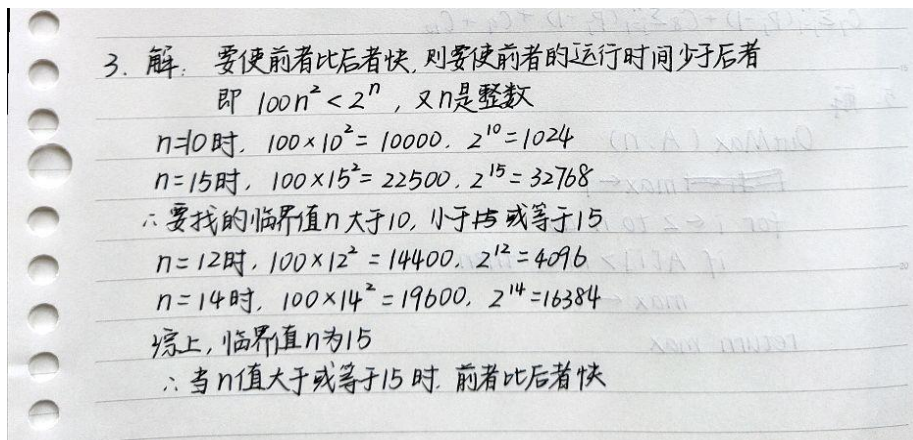
第一题



第二题



第三题



第四题

#4. 解:

	Compare(a, n)	Cost	times
1	j=1	C_1	1
2	while j<n and a[tj]<=a[tj+1] do	C_2	$\sum_{j=1}^n t_j$
3	j=j+1	C_3	$\sum_{j=1}^n (t_j-1)$
4	if j=n	C_4	$\sum_{j=1}^n (t_j-1)$
5	return true	C_5	1
6	while j<n and a[tj]>=a[tj+1] do	C_6	$\sum_{j=1}^n p_j$
7	j=j+1	C_7	$\sum_{j=1}^n (p_j-1)$
8	if j=n	C_8	$\sum_{j=1}^n (p_j-1)$
9	return true	C_9	1
10	else return false	C_{10}	1

$T(n) = C_1 + C_2 \sum_{j=1}^n t_j + C_3 \sum_{j=1}^n (t_j-1) + C_4 \sum_{j=1}^n (t_j-1) + C_5 + C_6 \sum_{j=1}^n (p_j-1) + C_7 \sum_{j=1}^n (p_j-1) + C_8 \sum_{j=1}^n (p_j-1) + C_9 + C_{10}$

第五题

```

J. MAX_Index(A)
  Max = A[0]; Max_index = 0;
  for j ← 1 to length(A)-1
    if (A[j] >= A[j-1])
      { Max = A[j];
        Max_index = j;
      }
  }
  return Max_index;

```